

13.1 Chain-of-Thought

CoT is extra tokens, not a new net. $17 \times 24 \rightarrow 408$ on the tape.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Chain-of-thought (CoT) is the habit of writing **steps** before the answer. You are not changing the weights. You are changing what the decoder is asked to emit.

First time this method appears. **Chain-of-thought (CoT)** is writing **steps** before the answer. It is a prompt / decoding choice, not a new net.

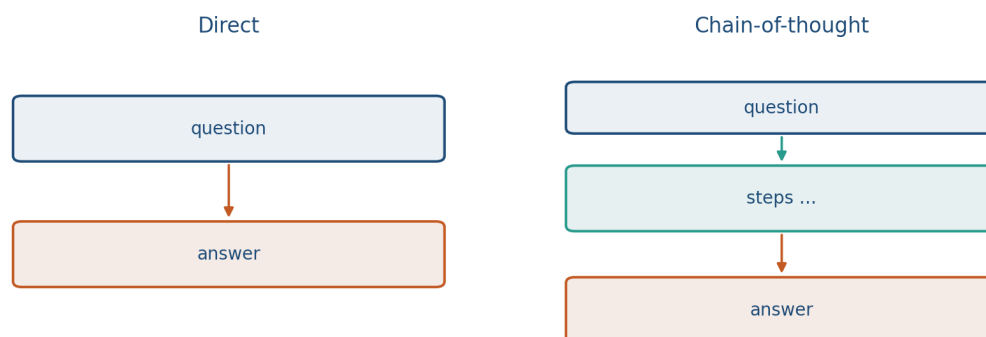
What. Direct: "What is 17×24 ?" CoT: "Show your work, then the number." Few-shot: paste worked traces. Zero-shot: *Let's think step by step.* **Why.** Extra tokens are extra test-time compute. Multi-hop arithmetic can use the tape. Weights do not change. **Architecture.** Same LM. Longer generation. Parse a final boxed answer separately from the trace (note 13.3). **How.** Freeze the instruction and decoding settings. Temperature 0 is one greedy chain, not a vote (13.2). **Formula.** Still

$\prod_t p(y_t | y_{<t}, \text{prompt})$; the prompt asks for steps. No new parameters. **Tradeoffs.** + Helps multi-hop. – Hurts one-hop lookup; latency/cost; fluent wrong algebra (faithfulness next). Not retrieval or tools.

1. Steps are part of the string

A direct prompt is "What is 17×24 ?" A CoT prompt is "Show your work, then give the number." Few-shot CoT puts worked examples in the context; zero-shot CoT is often the sentence *Let's think step by step.*

This is a **prompt / decoding** choice, not a new architecture. Extra tokens are extra **test-time compute**. The weights do not change.



The model still predicts tokens. Intermediate tokens can allocate compute: they store partial products, units, or a plan. That helps **multi-step** arithmetic, symbolic puzzles, and some school-science items. It

does not magically add retrieval (Week 14) or tools. A longer string is not “thinking” in the everyday sense; it is extra decoding you can inspect.

Karpathy’s phrase is System 2 as more tokens. Same idea: each token is a fixed-size compute step, so a hard problem needs a longer tape.

Zero-shot vs few-shot is a method choice. Few-shot CoT is only as good as the traces you paste: same task, same answer format, same units. A mismatch is not “the model cannot reason”; it is a context that points at the wrong procedure.

2. When it helps, when it does not

CoT tends to help when the gold answer needs several dependent hops. It can hurt on tasks that are a single lookup: extra steps are extra chances to wander. Longer traces cost latency and dollars. For a project, CoT is a **method knob**, not a personality.

You still need a metric on the **answer**, and a separate look at the trace (note **13.3**). A correct box after a wrong derivation is not the same system as a correct derivation. Self-consistency (note **13.2**) is what you do when one chain is too noisy to trust.

3. How to write it in a report

Say whether the prompt was zero-shot or few-shot, paste the exact instruction, and freeze decoding settings (temperature, max tokens). Do not claim the model “reasoned” unless you have a check on the steps. Lab 13 will grade traces as strings, with no API.

Temperature 0 is a single greedy chain: useful as a baseline, not as a vote (note **13.2**). If you raise temperature to get diverse traces, you must still parse a final answer the same way every time.

4. Teaching this note

About **30 minutes** at the board, then **~8 minutes** of video. First of three Week-13 notes; Lab 13 is constructed traces, not an API.

- **0–10 min.** Direct vs CoT prompt. Tokens as scratch paper.
 - **10–20 min.** When CoT helps (multi-hop) vs hurts (lookup). Cost.
 - **20–30 min.** Worked 17×24 on the board, then a one-hop counterexample.
 - **Then** play Karpathy intro **35:00–38:02** (Thinking, System 1/2). Pause on “models need tokens to think.”
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5. Worked example

Prompt: 17×24 . Direct decode might jump to a familiar number (400, 408, 428) with no check.

A CoT string that actually computes:

1. $10 \times 24 = 240$
2. $7 \times 24 = 168$
3. $240 + 168 = 408$

Three intermediate tokens (here, three lines) hold partial products. The boxed answer is 408. Gold is 408. This trace is also **faithful** (note **13.3**): the last computed value matches the box.

17×24 : three lines hold the partial products. Direct decode has no tape.

step	value
10×24	240
7×24	168
sum	408

A one-hop item, "Capital of France?", does not need hops. A CoT prompt can invent a story ("Lyon was the capital in...") and then still say Paris—or wander to Lyon. Extra decode is extra risk. That is why you report **answer accuracy** and, separately, a trace check, not a vibe that the model "showed work."

Lab 13's $23+19$ traces are the same algebra with smaller numbers.

6. Where students get stuck

- Calling CoT a fine-tune. It is a **prompt / decoding** choice unless you trained on traces.
- Scoring only the box. Wrong steps plus a lucky 408 will look like a win until note **13.3**.
- Pasting few-shot examples from a different task (math shots, then a policy question) and blaming the model.

7. Video

Watch [Andrej Karpathy, Intro to Large Language Models](#), **35:00–38:02**.

Pause on System 1 vs System 2: the model gets more compute by emitting more tokens. That is extra test-time compute in classroom language. Optional nearby: **38:02–40:45** (self-improvement) is not

required for this note. Notes **14.1–14.2** and **15.1** reuse this same URL at other minutes.

8. Practice

1. Why might CoT raise accuracy on a four-hop word problem and lower it on “What is the capital of France?”
2. Few-shot CoT puts examples in context. What goes wrong if those examples are a different task than the test item?
3. Name one number you would report besides final-answer accuracy if your project uses CoT.
4. Write a three-line CoT for 17×24 that ends in 408. Then write one line that is CoT-shaped but computes $17 \times 20 = 340$ and boxes 408. Which is unfaithful?
5. Direct prompt, temperature 0, one sample: accuracy 0.62. CoT, temperature 0, one sample: accuracy 0.71, mean tokens $4 \times$ longer. What two numbers belong in the table besides 0.71?